

COM 225 OPEN SOURCE SYSTEMS – PAST EXAMINATION REVISION & ANSWER KEY

SECTION A: COMPULSORY (30 MARKS) – QUESTION ONE

a) Using examples describe the following data types as used in PHP.

i) A variable

A variable in PHP is a container that stores data. It starts with a `$` sign, followed by the variable name. Variables are case-sensitive and do not require explicit type declaration.

Examples:

php

```
$name = "John Doe";    // String variable
$age = 25;              // Integer variable
$price = 19.99;         // Float variable
$isValid = true;        // Boolean variable
```

ii) Boolean

A Boolean data type represents only two possible values: `true` or `false`. It is commonly used in conditional statements and logical operations.

Examples:

php

```
$isLoggedIn = true;
$isAdmin = false;

if ($isLoggedIn) {
    echo "Welcome back!";
}
```

b) Write a PHP script that connects to a MySQL database and retrieves all records from a user's table.

php

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "my_database";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT * FROM users";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    while($row = $result->fetch_assoc()) {
        echo "ID: " . $row["id"] . " - Name: " . $row["name"] . " - Email: " . $row["email"] . "<br>";
    }
} else {
    echo "0 results";
}

$conn->close();
?>
```

c) Explain any three benefits a start-up firm can get by using open source software.

1. **Cost Savings** – Start-ups have limited budgets. OSS is free to download, use, and modify, eliminating expensive licensing fees.
2. **Flexibility & Customization** – Start-ups can modify the source code to fit their specific business needs without waiting for a vendor's permission.
3. **Avoid Vendor Lock-in** – OSS allows start-ups to switch service providers or hosting platforms easily, preventing dependency on a single commercial vendor.

d) Discuss the following terms as used in Open source systems.

i) Freeware

Freeware is software that is available for use at no cost, but its source code is **not** open. Users cannot modify or redistribute it. Examples: Adobe Acrobat Reader, Skype.

ii) Free software

Free software emphasizes **freedom** (not price). Users have the four essential freedoms: to run, study, modify, and redistribute the software. The source code must be available. Example: GNU/Linux.

iii) Shareware

Shareware is initially free to try, but continued use requires payment. It often has limited features or a trial period. Source code is not provided. Examples: WinRAR, many antivirus trials.

e) Write a PHP script that generates and prints the first 10 Fibonacci numbers.

php

```
<?php
$n = 10;
$a = 0;
$b = 1;

echo "First 10 Fibonacci numbers: ";
for ($i = 1; $i <= $n; $i++) {
    echo $a . " ";
    $next = $a + $b;
    $a = $b;
    $b = $next;
}
?>
```

f) Explain the purpose of the break and continue statements in loops? Provide an example of each.

Statement	Purpose
-----------	---------

break	Terminates the entire loop immediately, exiting the loop structure.
--------------	---

continue	Skips the current iteration and moves to the next iteration of the loop.
-----------------	--

Example of break:

php

```
for ($i = 1; $i <= 10; $i++) {
    if ($i == 5) {
```

```
        break; // Loop stops when i = 5
    }
    echo $i . " ";
}
// Output: 1 2 3 4
```

Example of continue:

php

```
for ($i = 1; $i <= 5; $i++) {
    if ($i == 3) {
        continue; // Skip number 3
    }
    echo $i . " ";
}
// Output: 1 2 4 5
```

g) Write a PHP script that takes a number as input and checks if it is positive, negative, or zero using an if-else statement.

php

```
<?php
$num = -5; // Example input

if ($num > 0) {
    echo "$num is positive.";
} elseif ($num < 0) {
    echo "$num is negative.";
} else {
    echo "The number is zero.";
}
?>
```

h) Discuss any three application areas of Open source system.

1. **Web Servers** – Apache HTTP Server and Nginx (both open source) power over 60% of websites worldwide.
2. **Operating Systems** – Linux (open source) runs on servers, supercomputers, Android phones, and embedded devices.
3. **Databases** – MySQL and PostgreSQL are open source RDBMS used by companies like Facebook, Twitter, and Uber for data storage and management.

i) Using an example code demonstrate the difference between a local and a global variable.

php

```
<?php
$globalVar = "I am global"; // Global variable

function test() {
    $localVar = "I am local"; // Local variable
    echo $localVar . "<br>";

    // To access global variable inside function, use 'global' keyword
    global $globalVar;
    echo $globalVar . "<br>";
}

test();

// Access outside function
echo $globalVar . "<br>"; // Works
// echo $localVar;      // ERROR: undefined variable
?>
```

Difference:

- **Global variable** – declared outside any function, accessible everywhere.
- **Local variable** – declared inside a function, accessible only within that function.

j) Write a PHP program using a switch statement that takes a number (1-7) as input and prints the corresponding day of the week.

php

```
<?php
$dayNumber = 3; // Example input

switch ($dayNumber) {
    case 1:
        echo "Monday";
        break;
    case 2:
        echo "Tuesday";
        break;
    case 3:
        echo "Wednesday";
        break;
    case 4:
        echo "Thursday";
        break;
    case 5:
        echo "Friday";
        break;
    case 6:
        echo "Saturday";
        break;
    case 7:
        echo "Sunday";
        break;
    default:
        echo "Invalid number. Please enter 1-7.";
```

```
}  
?>
```

k) Write a PHP script using a for loop to print numbers from 1 to 10.

php

```
<?php  
for ($i = 1; $i <= 10; $i++) {  
    echo $i . " ";  
}  
// Output: 1 2 3 4 5 6 7 8 9 10  
?>
```

l) Explain what an Index is in MySQL and how it improves performance.

Index is a database structure that improves the speed of data retrieval operations (SELECT queries). It works like an index in a book – instead of scanning every row, MySQL uses the index to quickly locate the data.

How it improves performance:

- Reduces the number of disk I/O operations.
- Enables fast searching, sorting, and grouping.
- Without an index, MySQL performs a **full table scan** (slow for large tables).

Example:

sql

```
CREATE INDEX idx_lastname ON employees(last_name);
```


m) A company needs to manage database security and prevent unauthorized access.

Key measures:

1. **User Authentication** – Enforce strong passwords and use `CREATE USER` with `IDENTIFIED BY`.
2. **Privilege Management** – Grant only necessary privileges (e.g., `SELECT`, `INSERT`) using `GRANT` and `REVOKE`.
3. **Encryption** – Use SSL/TLS for data in transit and encrypt sensitive columns (e.g., `AES_ENCRYPT()`).
4. **Regular Audits** – Monitor logs and use `SHOW GRANTS` to review user permissions.
5. **Firewall & Network Security** – Restrict MySQL port (3306) to trusted IPs only.
6. **Backup & Recovery** – Regular automated backups to prevent data loss.

SECTION B: (40 MARKS) – CHOOSE ANY TWO QUESTIONS

Note: For revision, we answer all four questions (Two, Four, Five). Missing Question Three (maybe privileges? but we have privileges in Four? Actually Four i) is privileges).

QUESTION TWO (20 MARKS)

(Many sub-questions repeat Section A – see answers above for a,c,d,e,f,g)

b) Discuss any three application areas of Open source system. – See Section A (h)

g) A company needs to manage database security... – See Section A (m)

QUESTION FOUR (20 MARKS)

a) Write a PHP script that prints the multiplication table of 5 using a while loop.

php

```
<?php
$num = 5;
$i = 1;

while ($i <= 10) {
    echo "$num x $i = " . ($num * $i) . "<br>";
    $i++;
}
?>
```

b) Employees database queries

i) Insert a new employee:

sql

```
INSERT INTO employees (empId, Name, Age, Department, Salary, JoiningDate)
VALUES (101, 'John Otieno', 30, 'IT', 75000.00, '2023-06-15');
```

ii) Update salary of IT department by 10%:

sql

```
UPDATE employees
SET Salary = Salary * 1.10
WHERE Department = 'IT';
```

iii) Delete employees who have worked more than 10 years:

sql

```
DELETE FROM employees
WHERE JoiningDate <= CURDATE() - INTERVAL 10 YEAR;
```

Alternative for older MySQL: `WHERE DATEDIFF(CURDATE(), JoiningDate) > 3650`

c) PHP script demonstrating a recursive function

Example: Factorial calculation

php

```
<?php
function factorial($n) {
    if ($n <= 1) {
        return 1;
    } else {
        return $n * factorial($n - 1);
    }
}

echo "Factorial of 5 is: " . factorial(5); // Output: 120
?>
```

i) Explain different types of privileges in MySQL (8 marks)

MySQL privileges are categorized as:

Privilege Level	Description	Examples
Global	Apply to all databases on the server	<code>CREATE USER</code> , <code>PROCESS</code> , <code>SHUTDOWN</code> , <code>SUPER</code>

Privilege Level	Description	Examples
Database	Apply to all objects within a specific database	<code>CREATE, DROP, ALTER</code> on <code>db.*</code>
Table	Apply to specific tables	<code>SELECT, INSERT, UPDATE, DELETE</code> on <code>db.table</code>
Column	Apply to specific columns of a table	<code>SELECT(col1), UPDATE(col2)</code>
Stored Routine	Apply to functions/procedures	<code>EXECUTE, ALTER ROUTINE</code>

Key privileges:

- `ALL PRIVILEGES` – Grants all available permissions.
- `SELECT` – Read data.
- `INSERT` – Add new rows.
- `UPDATE` – Modify existing data.
- `DELETE` – Remove rows.
- `CREATE / DROP` – Create or delete databases/tables.
- `INDEX` – Create or drop indexes.
- `GRANT OPTION` – Allows user to grant privileges to others.

Example:

sql

```
GRANT SELECT, INSERT ON company.* TO 'manager'@'localhost' IDENTIFIED BY 'securepass'
;
REVOKE DELETE ON company.* FROM 'manager'@'localhost';
```

QUESTION FIVE (20 MARKS)

a) Trends and potential of Free Open Source Systems

Trends:

1. **Cloud-native adoption** – Kubernetes, Docker, Prometheus are open source standards.
2. **AI/ML democratization** – TensorFlow, PyTorch, Scikit-learn enable accessible AI.
3. **Edge computing & IoT** – Open source OS (Linux, Android Things) dominate.
4. **Cybersecurity** – Open source tools (Snort, Wireshark, Metasploit) lead the field.
5. **Government adoption** – Many countries mandate OSS for transparency (e.g., India, EU).

Potential:

- Reduced digital divide in developing nations.
- Faster innovation through global collaboration.
- Long-term sustainability of critical software.
- Empowerment of start-ups and small businesses.

b) Two dangers of using Freeware and Shareware

1. **Malware & Spyware** – Freeware/shareware may bundle adware, trackers, or actual malware since the source code is not open for review. Users unknowingly install keyloggers or data harvesters.
2. **Lack of Updates & Support** – Developers of free/shareware have no obligation to provide security patches. Exploits can remain unpatched, leaving systems vulnerable to attacks (e.g., old versions of Java or PDF readers).

c) SQL vs NoSQL comparison

i) Key differences between SQL and NoSQL databases

Feature	SQL (Relational)	NoSQL (e.g., MongoDB)
Data model	Tables with rows and columns	Documents (JSON), key-value, graphs, wide-column
Schema	Fixed, predefined schema	Dynamic, schema-less
ACID compliance	Strong ACID transactions	Eventual consistency (BASE)
Scalability	Vertical (scale-up)	Horizontal (scale-out)
Query language	SQL (structured)	API-based or proprietary
Joins	Supported	Limited or denormalization required

ii) Advantages of NoSQL compared to SQL

- **Horizontal scalability** – Distributes data across cheap commodity servers easily.
- **High write throughput** – Handles massive volumes of writes (e.g., logs, IoT data).
- **Flexible schema** – Accommodates evolving data structures without migrations.
- **Performance** – Faster for simple read/write patterns with large data volumes.
- **Native support for unstructured/semi-structured data** (JSON, XML).

iii) Real-world scenarios where each is better

NoSQL is better for:

- **Real-time analytics** (e.g., user activity logs, clickstream data)
- **Content management systems** with diverse content types (e.g., blog platforms)
- **IoT data ingestion** (time-series data, high-velocity writes)
- **Social networks** (graph databases for friend relationships)

SQL is better for:

- **Banking/financial systems** requiring strict ACID transactions
- **ERP or CRM systems** with complex relationships and reporting
- **E-commerce order management** (inventory, orders, payments)
- **Any application where data integrity and joins are critical**

PREDICTION FOR THIS YEAR'S EXAM (2025/2026)

Based on the past paper, the course outline, and common topics, expect the following:

High Probability Topics (90%+)

Topic

Expected Question Type

PHP basics (variables, loops, functions, arrays)

Write scripts (Fibonacci, factorial, multiplication table, prime numbers, palindrome)

MySQL CRUD (INSERT, SELECT, UPDATE, DELETE)

Write SQL queries + PHP integration

Topic	Expected Question Type
Open Source benefits & challenges	Short essays (5 marks each)
Licensing (GPL, MIT, Apache, Freeware, Shareware)	Definitions and comparisons
Linux basics (commands, file permissions, kernel vs user mode)	Short answers or practical commands
Process management (fork, signals, scheduling)	Theory questions (4-8 marks)
Database indexes & privileges	Explanation + examples

Medium Probability (60-70%)

- **SQL vs NoSQL** differences (as seen in last year)
- **Recursive functions** in PHP
- **Break / continue** examples
- **Local vs global variables**
- **Open source application areas** (e-commerce, education, government)
- **Vendor lock-in** concept

Low Probability but Possible

- Detailed Linux architecture diagrams
- Clone() system call details
- Full stack development (LAMP stack setup)

- Eclipse IDE usage

Practical Scripts to Practice

1. Fibonacci series (iterative & recursive)
2. Prime number checker
3. Factorial using recursion
4. Array sorting (bubble sort, merge sort)
5. Login system with session
6. File upload and read
7. MySQL connection and display data in HTML table
8. Form validation with PHP

Sample Predicted Question

"Write a PHP script that accepts a number from a form, checks whether it is a prime number, and displays the result."

Model answer:

php

```
<?php
function isPrime($n) {
    if ($n <= 1) return false;
    for ($i = 2; $i <= sqrt($n); $i++) {
        if ($n % $i == 0) return false;
    }
    return true;
}

$num = $_POST['number'] ?? 0;
if (isPrime($num)) {
    echo "$num is a prime number.";
} else {
    echo "$num is not prime.";
```

```
}  
?>
```

Best of luck in your exam! Revise all practical scripts, MySQL queries, and open source definitions.